

Pure ReLU Networks Already Give Compact-Support Universality

A literature-implied full answer to the pooling question in arXiv:2204.11231

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Source question

Kratsios and Zamanlooy construct ReLU networks with bilinear pooling that approximate compactly supported functions while controlling the support [1]. On PDF page 16 they explain that replacing the exact product by a ReLU approximation to multiplication need not preserve compact support. They therefore leave open whether one can construct a deep ReLU network without pooling that satisfies the hypotheses of their Lemma 2.

That lemma asks for the following. Given a compactly supported Lipschitz map $f : \mathbb{R}^d \rightarrow \mathbb{R}^D$, find approximants f_m such that

$$\|f_m - f\|_{L^1(\mathbb{R}^d; \mathbb{R}^D)} \rightarrow 0$$

and such that $\text{ess-supp } f$ and all $\text{ess-supp } f_m$ lie in one fixed enlarged cube. Under these two conditions, Lemma 2 concludes that $f_m \rightarrow f$ in the paper's compactly-supported L^1 topology τ .

Result

Let $\mathcal{N}_{d,D}^{\text{ReLU}}$ denote the functions $\mathbb{R}^d \rightarrow \mathbb{R}^D$ realized by finite feedforward ReLU networks with affine output layer.

Theorem 1 (Pooling is unnecessary). *Let $d, D \geq 1$, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}^D$ be Lipschitz with compact essential support. If*

$$n_f = \min\{n \in \mathbb{N} : \text{ess-supp } f \subseteq [-n, n]^d\}, \quad Q = [-n_f - 1, n_f + 1]^d,$$

then for every $\varepsilon > 0$ there is $N_\varepsilon \in \mathcal{N}_{d,D}^{\text{ReLU}}$ such that

$$\|N_\varepsilon - f\|_{L^\infty(\mathbb{R}^d; \mathbb{R}^D)} < \varepsilon, \quad \text{ess-supp } N_\varepsilon \subseteq Q.$$

Consequently, for the csL^1 -topology τ of arXiv:2204.11231,

$$\overline{\mathcal{N}_{d,D}^{\text{ReLU}}}^\tau = L_{\text{loc}}^1(\mathbb{R}^d; \mathbb{R}^D).$$

Idea of the proof

The product-and-mask route is not needed. On a simplicial mesh of the fixed cube Q , interpolate f at the vertices. The interpolation error is bounded by the Lipschitz constant times the mesh diameter. Because the cube has a full collar outside the support of f , all boundary data are zero; the interpolant therefore extends by zero to a global compactly supported continuous piecewise-affine map. A pre-existing exact representation theorem then turns this finite CPWL map into a finite ReLU network without changing it or its support.

Proof

For a continuous function, essential support agrees with the closure of its nonzero set. Indeed, if $f(x) \neq 0$, continuity makes f nonzero on a set of positive measure near x . Hence $f = 0$ outside $[-n_f, n_f]^d$, and in particular on a neighbourhood of ∂Q .

Write $L = \text{Lip}(f)$. If $L = 0$, compact support forces $f = 0$, and the zero network proves the claim. Assume $L > 0$. Choose a finite conforming simplicial triangulation $\mathcal{T}_h$ of Q whose mesh diameter

$$h := \max_{S \in \mathcal{T}_h} \text{diam}(S)$$

is smaller than ε/L . Let $I_h f$ be the nodal affine interpolant of f on this triangulation. If x belongs to a simplex S with vertices $v_0, \dots, v_d$ and barycentric coordinates $\lambda_0, \dots, \lambda_d$, then

$$I_h f(x) = \sum_{i=0}^d \lambda_i f(v_i).$$

Therefore

$$\begin{aligned} \|I_h f(x) - f(x)\| &\leq \sum_{i=0}^d \lambda_i \|f(v_i) - f(x)\| \\ &\leq L \sum_{i=0}^d \lambda_i \|v_i - x\| \leq Lh < \varepsilon. \end{aligned}$$

Every boundary vertex of Q has value zero. On each boundary face the interpolant is affine with zero vertex values, so its boundary trace is zero. Extend $I_h f$ by zero on $\mathbb{R}^d \setminus Q$, and call the resulting map g_h . It is globally continuous, affine on each member of a finite polyhedral decomposition of $\mathbb{R}^d$, and satisfies

$$\text{supp } g_h \subseteq Q, \quad \|g_h - f\|_\infty < \varepsilon.$$

Thus every coordinate of g_h is a finite scalar CPWL function.

Theorem 5.2 on PDF page 23 of He–Li–Xu–Zheng states that every finite scalar CPWL function on $\mathbb{R}^d$ is represented exactly by a finite ReLU DNN [2]. Apply that theorem to the D coordinates of g_h , run the resulting networks in parallel, and collect their outputs with one affine output layer. The resulting pure ReLU network N_ε equals g_h pointwise. It therefore has the claimed uniform error and support.

Choose $h_m \downarrow 0$. The networks just constructed have all their supports in the same cube Q , and

$$\|N_m - f\|_{L^1} \leq |Q| \|N_m - f\|_\infty \longrightarrow 0.$$

These are exactly the two hypotheses of Lemma 2 of the source paper (PDF page 12), so $N_m \rightarrow f$ in τ . Lemma 1 of that paper says that compactly supported Lipschitz maps are τ -dense in L^1_{loc} . The usual transitivity-of-density argument used in the source now gives the asserted τ -density of pure ReLU networks. $\square$

Status and scope

This completely answers the qualitative question on PDF page 16: a pure ReLU network can meet Lemma 2, so bilinear pooling is not required for $\text{cs}L^1$ universality. The result also gives uniform approximation of each compactly supported Lipschitz target with support in the fixed one-unit

enlargement Q . It does not claim the specific width, depth, or metric-capacity bounds of the source’s quantitative Theorem 3.

The status is `literature_implied_answer`: the interpolation reduction above identifies the answer, while the decisive exact CPWL-to-ReLU representation theorem was already available before the source paper. A search of the run indexes and arXiv/web queries found no paper explicitly presenting this reduction as an answer to the question in arXiv:2204.11231.

References

- [1] A. Kratsios and B. Zamanlooy, *Do ReLU Networks Have An Edge When Approximating Compactly-Supported Functions?*, Transactions on Machine Learning Research (2022), arXiv:2204.11231.
- [2] J. He, L. Li, J. Xu, and C. Zheng, *ReLU Deep Neural Networks and Linear Finite Elements*, arXiv:1807.03973, Theorem 5.2, PDF p. 23 (2018). The paper records that the exact general CPWL representation theorem is due to R. Arora, A. Basu, P. Mianjy, and A. Mukherjee, *Understanding Deep Neural Networks with Rectified Linear Units*, arXiv:1611.01491.